Day 2 Exercises

Forest type classification and carbon. The carbon section contains a trap that has been deliberately left in — finding it is the exercise.

A Classification concepts

A1

Explain in two sentences why you cannot separate evergreen from deciduous forest using a single Sentinel-2 image, no matter how clear it is.

A2

For each forest type, say whether NDVI amplitude (wet minus dry) is high or low, and why.

FOREST TYPE	AMPLITUDE	REASON
Evergreen (ป่าดิบ)		
Deciduous (ป่าผลัดใบ)		
Rubber plantation		

A3

A colleague reports a forest type map with 99.4% overall accuracy. Should you be impressed? Give the most likely explanation.

A4

Here is a confusion matrix from a four-class map. Rows are truth, columns are prediction.

	EVERGREEN	DECIDUOUS	CROPLAND	WATER
Evergreen	79	0	2	2
Deciduous	0	73	8	2
Cropland	4	10	51	8
Water	0	1	10	92

(a) Which class is classified **worst**? (b) Which two classes are confused most often? (c) Give a physical reason why.

A5

Why must the confusion matrix be computed on points the model has **never seen**? What specifically goes wrong if you use the training points?

B Carbon arithmetic

B1

A forest plot has an aboveground biomass of **240 Mg/ha**. Work through the chain. Show each step.

STEP	VALUE
Aboveground biomass	240 Mg/ha
Aboveground **carbon** (× 0.47)	
CO_2 equivalent per hectare (× 44/12)	
Total CO_2e over 500 ha	

B2 · THE TRAP

A student writes this and reports **21.7 Mg C/ha** for a forest that everyone agrees is healthy. What is wrong, and what is the correct answer?

```
ornl = ee.ImageCollection('NASA/ORNL/biomass_carbon_density/v1').first()
agb = ornl.select('agb')
carbon = agb.multiply(0.47)      # convert biomass to carbon
```

B3

Fill in whether each dataset needs the 0.47 carbon fraction applied.

DATASET AND BAND	UNITS	APPLY × 0.47?
`ORNL ... /v1` band `agb`		
`GEDI04_B_002` band `MU`		

B4

Your reduceRegion returns a total carbon stock of **1,847** tonnes for a whole province of forest. Is that plausible? If not, what mistake produces a number that small?

C Lab work

C1 · Lab 3 — classify your province

QUANTITY	YOUR VALUE
Province	
NDVI amplitude, median over tree cover	
Number of training points	
Overall accuracy	
Kappa	
Evergreen forest (ha)	
Deciduous forest (ha)	

Sniff test. Is your accuracy between 0.70 and 0.95? ☐ yes ☐ no — *if above 0.98, suspect a leak.*

C2 · Does the method suit your province?

Our thresholds were amplitude < 0.08 for evergreen and > 0.20 for deciduous. Look at your own percentiles. Are those thresholds sensible for your province, or would you move them? Justify with your numbers.

C3 · Sample size

Re-run Lab 3 with `numPoints=50` instead of 300.

NUMPOINTS	OVERALL ACCURACY	KAPPA
300		
50		

What happened, and what does it tell you about how much training data a classifier needs?

C4 · Lab 4 — carbon in your province

QUANTITY	YOUR VALUE
ORNL mean aboveground (Mg C/ha)	
ORNL mean belowground (Mg C/ha)	
Total carbon stock (t C)	
Total CO_2 equivalent (t CO_2e)	
GEDI mean biomass (Mg/ha)	
GEDI converted to carbon (Mg C/ha)	
Ratio GEDI : ORNL	

C5 · Report it honestly

Write the carbon density of your province as a **range with both sources and their years**, in one sentence a reviewer could not object to.

D Homework

D1

Your province's forest is being considered for a carbon credit project. A consultant sends you a report claiming a single figure of "62.4 Mg C/ha, derived from satellite data". Write three questions you would ask them before accepting that number.

D2

Write a PROMPT-recipe prompt for something we have not done: *"map areas where forest was lost between 2015 and 2020"*, using the Hansen dataset. Fill all six slots. Pay attention to how `lossyear` is encoded — check the cheat sheet.